2D Matrices: Reflections

Questions 1–14

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Question 2. It is given that the matrix $\mathbf{A}$ represents a reflection in the y-axis.

a) Write down the matrix $\mathbf{A}$.

b) Find $\mathbf{A}^2$.

c) Find $\mathbf{A}^3$.

d) Find $\mathbf{A}^{2025}$.

Question 3. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Write down the area of the shape's image.

Question 4. Write down the matrix that represents a reflection that maps:

a) $(3, 5)$ to $(3, -5)$
b) $(-6, -2)$ to $(6, -2)$
c) $(1, 9)$ to $(-9, -1)$
d) $(-4, -7)$ to $(-7, -4)$

Question 5. The triangle T has vertices at $(8, 1)$, $(9, 3)$ and $(10, -3)$. T is reflected in the line $y = x$ to form T'. Using matrices, find the coordinates of the vertices of T'.

Question 6. The trapezium X has vertices at $(-1, 5)$, $(6, 5)$, $(11, -3)$ and $(-1, -3)$. X is reflected in the x-axis to form X'. Using matrices, find the coordinates of the vertices of X'.

Question 7. Use a matrix method to reflect the point $(3, -5)$ in the line with equation:

a) $y = x + 5$

b) $y = 8 - x$

Question 8. Write down the 2×2 matrix that represents:

a) A reflection in the line that makes an angle of $30°$ with the horizontal.

b) A reflection in the line that makes an angle of $60°$ with the horizontal.

c) A reflection in the line that makes an angle of $15°$ with the horizontal.

d) A reflection in the line that makes an angle of $75°$ with the horizontal.

Question 9. Write down the matrix that represents the following linear transformations:

a) A reflection in the line $y = \sqrt{2}x$.
b) A reflection in the line $y = 3x$.
c) A reflection in the line $y = \frac{1}{8}x$.
d) A reflection in the line $y = \frac{1}{\sqrt{3}}x$.
e) A reflection in the line $y = -2x$.
f) A reflection in the line $y = -\frac{x}{\sqrt{2}}$.
g) A reflection in the line $y = -\frac{1}{5}x$.
h) A reflection in the line $y = -\frac{x}{5\sqrt{3}}$.

Question 10. Find the equation of the mirror line for each of these matrices in the form $y = kx$, where k is given to 3 significant figures:

a) $\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$
b) $\begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$
c) $\begin{pmatrix} \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} \\ \frac{\sqrt{6}-\sqrt{2}}{4} & -\frac{\sqrt{6}+\sqrt{2}}{4} \end{pmatrix}$
d) $\begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} \\ \frac{\sqrt{6}+\sqrt{2}}{4} & -\frac{\sqrt{6}-\sqrt{2}}{4} \end{pmatrix}$

Question 11. The point $(4, 6)$ is mapped to the point $(3 + 2\sqrt{3}, 2 - 3\sqrt{3})$ by a reflection in the line $y = kx$. Find the matrix that represents this transformation.

Question 12. The point $(-18, 20)$ is mapped to the point $(9 + 10\sqrt{3}, 10 - 9\sqrt{3})$ by a reflection in the line $y = kx$. Find the matrix that represents this transformation.

Question 13. The 2×2 matrix $\mathbf{A}$ represents a reflection in the line $y = 4x$. Find the matrix $\mathbf{A}$, writing each element in an exact form not involving trigonometry.

Question 14. The matrix $\mathbf{A}$ represents a reflection in the line $y = kx$, where k is a constant. Show that $\mathbf{A} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}$.